

APIARY

Designing a User-Friendly Cybersecurity Management Platform

Duke | Code<+>
2025

**Blocklist & Safelist
Management Interface**

▶ 13:28



Blocklist & Safelist Management Interface

Develop an intuitive, web-based front-end for Apiary, a security platform managing cybersecurity threat intelligence. The system streamlines tasks like entry...



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Made with GAMMA

What is APIARY?

APIARY is a front-end application designed to manage blocklists, safelists, collections, inputs, and outputs within Duke University's cybersecurity system. Our objective was to create an intuitive interface that simplifies complex cybersecurity workflows for everyday security staff.

Blocklists & Safelists

Manage allow and deny lists for threat intelligence

Collections

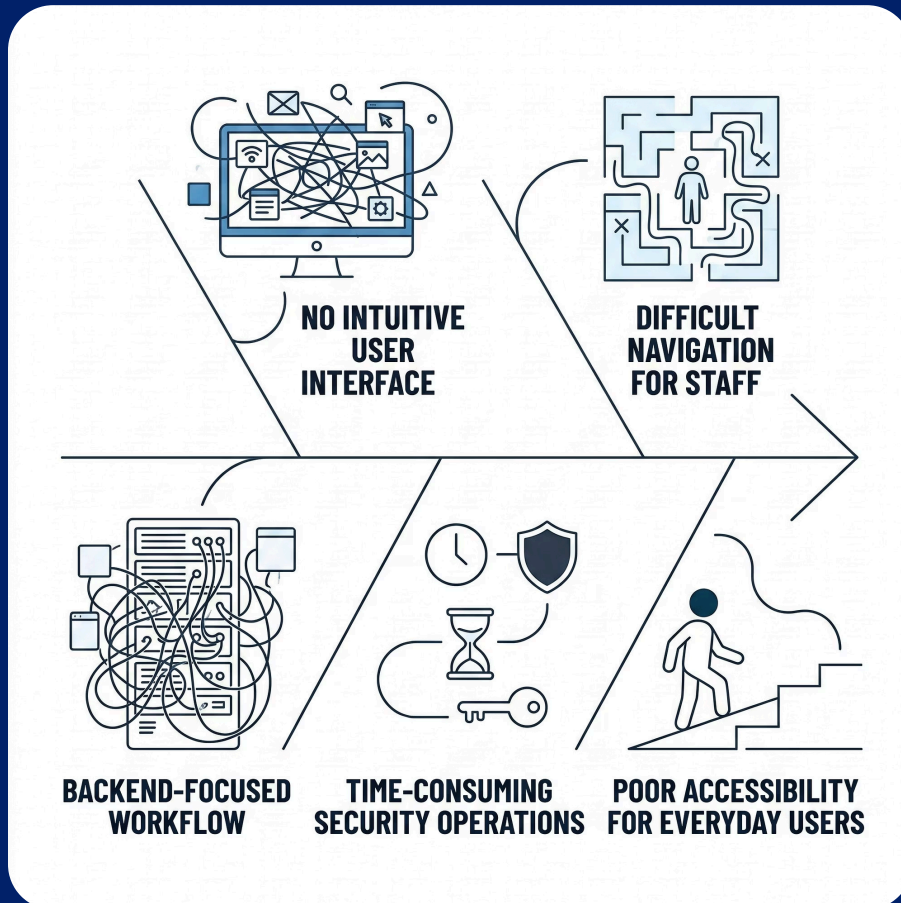
Organize intelligence into structured, searchable groups

Inputs & Outputs

Generate and track data flows across the security system

Existing System Challenges

The existing cybersecurity platform already had a functioning backend, but it lacked a usable front-end interface. Security staff struggled with a backend-focused workflow that was time-consuming and poorly accessible.



Design Challenge: How might we design an interface that allows users to quickly and confidently manage cybersecurity actions?

Stakeholder Collaboration

Throughout the project, we worked closely with stakeholders whose feedback guided every major design decision.



Anthony – Project Stakeholder

Primary point of contact for requirements and design validation



Duke OIT

Provided institutional context and operational constraints



Project Leaders & End Users

Security staff who would manage cybersecurity data day-to-day

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What Did Users Need?

From discussions with our stakeholder, we identified core requirements that became the foundation of our information architecture.

1

View Recent Activity

Monitor security events at a glance

2

Manage Blocklists & Safelists

Add, edit, and remove entries efficiently

3

Create & Edit Actions

Define and modify security response actions

4

Organize Collections

Structure threat intelligence into logical groups

5

Generate Inputs & Outputs

Produce and consume data across the system

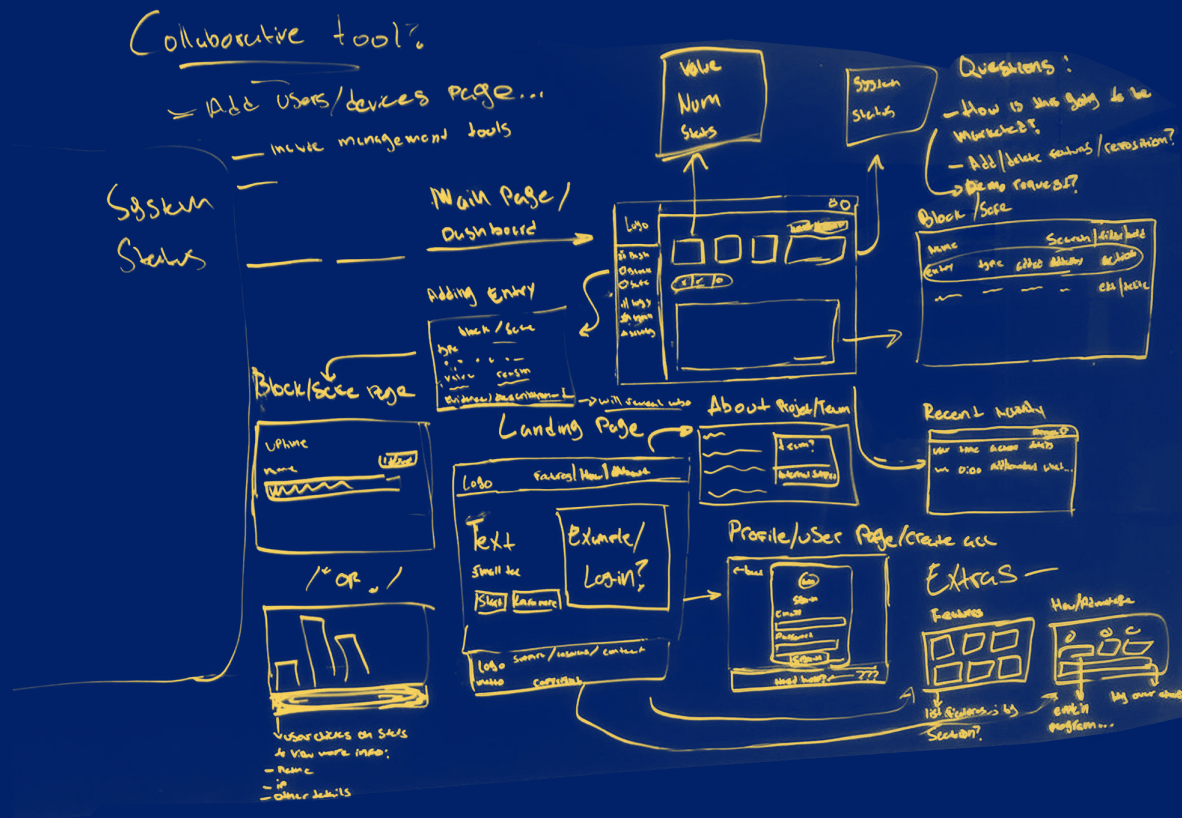
6

Navigate Easily

Move between pages without friction

Early Ideation

Before opening Figma, we mapped out every screen on a whiteboard – brainstorming all major pages and planning how users would move between them.



Screens Mapped

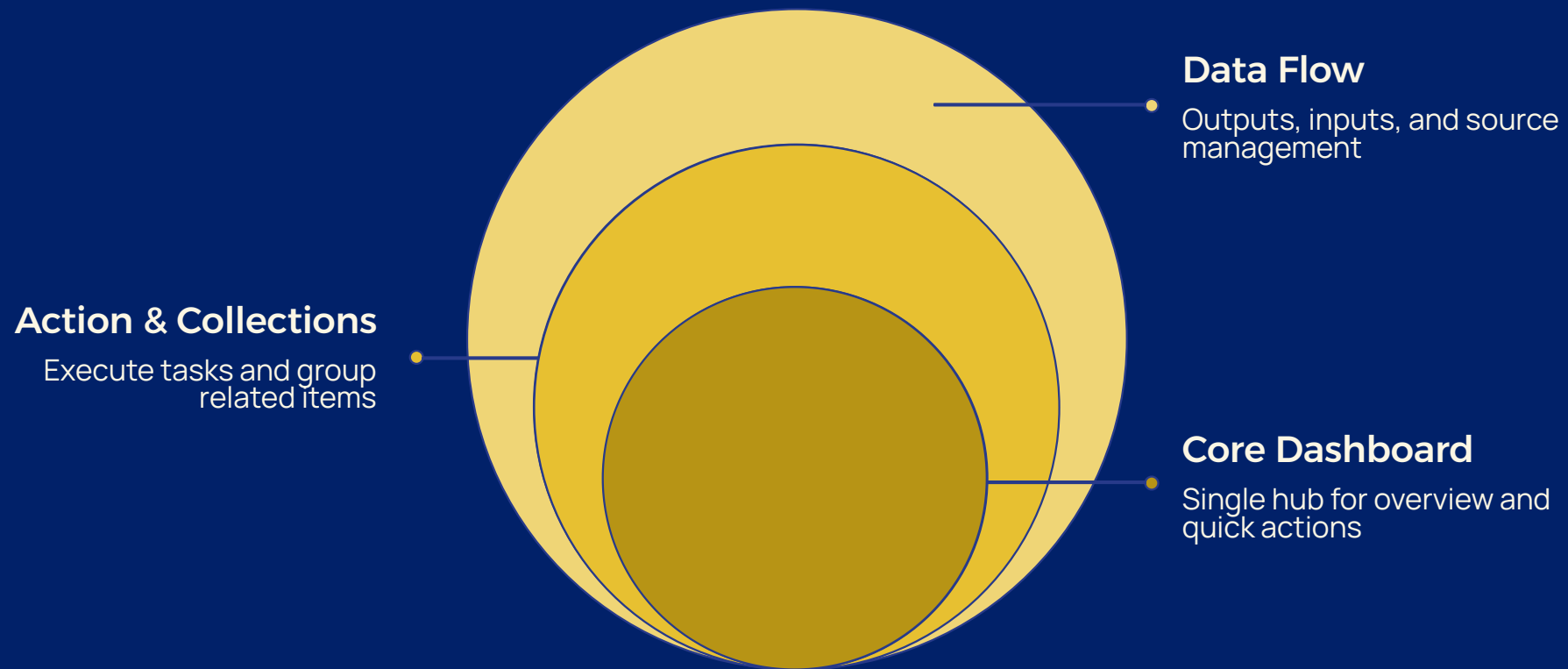
- Dashboard & Login
- Profile
- Blocklists & Safelists
- Collections & Sources
- Inputs & Outputs

Key Questions

- How do we make this easy to use?
- What should users see first?
- How can complex actions feel simple?
- How can we improve accessibility?

Organizing the Experience

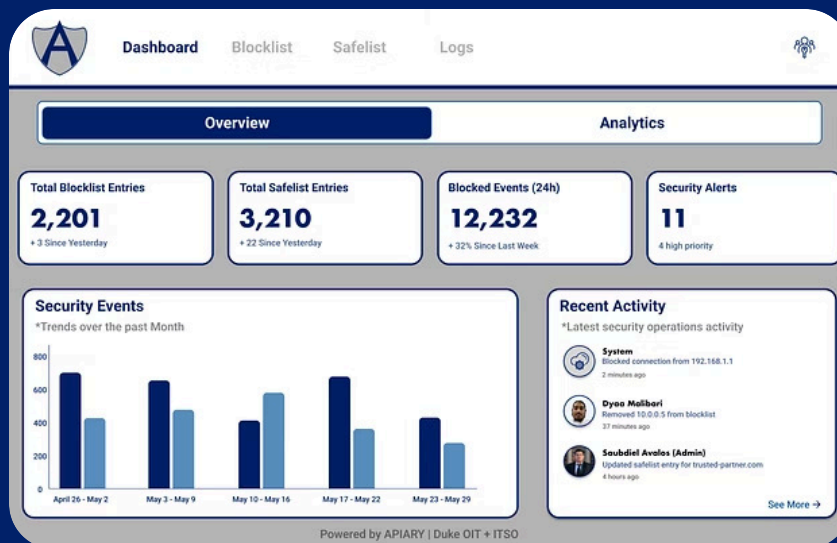
From brainstorming, we created the application's navigation hierarchy. Our goal was **minimizing clicks** while keeping related tasks grouped together.



This linear structure ensures users can reach any major section in just a few clicks, reducing cognitive load during high-pressure security operations.

First Prototype

Using our sketches, we created low-fidelity wireframes in Figma. Our focus was on structure and flow — not visual polish.



The Blocklist Management wireframe features a top navigation bar with links to Dashboard, Blocklist (selected), Safelist, and Logs. Below this, there are buttons for Export, Import, and Add Entry. A search bar is present for searching blocklist entries. The main content is a table with columns: IP/Domain, Type, Reason, Severity, Date Added, Added By, and Actions. The table lists several entries with details on their type, reason, severity, and who added them.

IP/Domain	Type	Reason	Severity	Date Added	Added By	Actions
192.168.1.1	IP	Malicious Activity	High	2023-05-26	System	Edit Remove
malware.example.com	Domain	Suspicious Behavior	Medium	2023-05-25	Saibidil Avalos (Admin)	Edit Remove
10.0.0.5	IP	Known malware distribution	Critical	2023-05-24	Saibidil Avalos (Admin)	Edit Remove
172.16.0.1	IP	Brute force attempts	High	2023-05-24	System	Edit Remove
phishing@example.net	Domain	Phishing campaign	Low	2023-05-22	Dyaa Malbari	Edit Remove
184.773.9.6	IP	Malicious Activity	Medium	2023-05-22	Shavit Kimchi	Edit Remove

Our Goals

- **Layout**
Define spatial organization of each screen
- **User Flow**
Map how users complete key tasks

- **Navigation**
Establish clear paths between sections
- **Components**
Place UI elements where they make sense

Why We Designed It This Way

Our first prototype reflected deliberate choices — balancing Duke's brand identity with the flexibility needed for future open-source adoption.



Duke Color Palette

Aligned with institutional branding for familiarity



Top Navigation

Standard pattern reducing learning curve



Dashboard Overview

At-a-glance visibility into recent security activity



Open-Source Ready

Flexible structure designed for broader adoption

What We Learned

After presenting our prototype, stakeholders identified several issues that shaped our next iteration. The most critical insight: **users wanted a darker workspace.**

Color Scheme

The light interface wasn't preferred. Security analysts work in low-light environments and requested a dark theme for reduced eye strain.

Information Hierarchy

Stakeholders clarified which data should be most prominent on the dashboard to support faster decision-making.

Workflow Efficiency

Feedback revealed opportunities to reduce steps for common tasks like adding blocklist entries and managing collections.

Design Iteration

Applying Feedback

We redesigned the interface based on stakeholder input, focusing on crucial improvements identified during the prototyping phase.



Dark mode

Implemented a dark theme for reduced eye strain during extended use.



Better accessibility

Ensured compliance with accessibility standards for diverse users.



Improved contrast

Optimized color combinations for readability in various lighting conditions.



Cleaner navigation

Streamlined menu structures for intuitive user journeys.



Stronger hierarchy

Clearly distinguished primary and secondary information on each screen.



More scalable layout

Designed for future expansion and integration of new features without clutter.

Branding Evolution

Building an Identity

We redesigned the visual identity around the concept of an apiary, incorporating elements that reflect the core values of cybersecurity management.



Shield-shaped "A"

This iconic element provides instant recognition and symbolizes the robust security and protection offered by the platform.



Honeycomb Elements

Representing structure and efficiency, these geometric patterns signify the organized and collaborative nature of threat intelligence.



Bee Illustration

The diligent bee embodies collaboration and collective effort, highlighting how various security components work together seamlessly.

Navigation Improvements

A direct comparison of our navigation redesign highlights key user experience enhancements, prioritizing clarity and ease of use.

Before: Initial Prototype

- Top navigation
- Less organization
- Crowded layout

After: Refined Design

- Sidebar navigation
- Clear page hierarchy
- More scalable
- Easier navigation

The shift to a sidebar navigation dramatically improved information architecture, making the platform feel less cluttered and more intuitive for daily operations.

Dashboard Improvements

Refining the Dashboard

The dashboard remained an important feature, providing users with essential elements for effective cybersecurity management:



Recent Activity

Monitor all critical security events at a glance.



System Overview

Understand the overall health and status of the system.



Quick Statistics

Access key metrics and performance indicators rapidly.



Faster Workflows

Streamlined processes for more efficient task completion.

We reorganized the layout to improve readability without changing its core functionality, ensuring critical information is always accessible.

Design Principles

UX Decisions



Simplicity

Reduced unnecessary complexity, making tasks straightforward.



Recognition

Created familiar navigation patterns to reduce learning curves.



Accessibility

Improved readability and contrast, ensuring usability for all users.



Efficiency

Reduced the number of steps required for common, critical tasks.

Development

From Design to Development

With a clear design vision established, our team transitioned to the development phase, leveraging modern technologies to build a robust and scalable cybersecurity management platform.

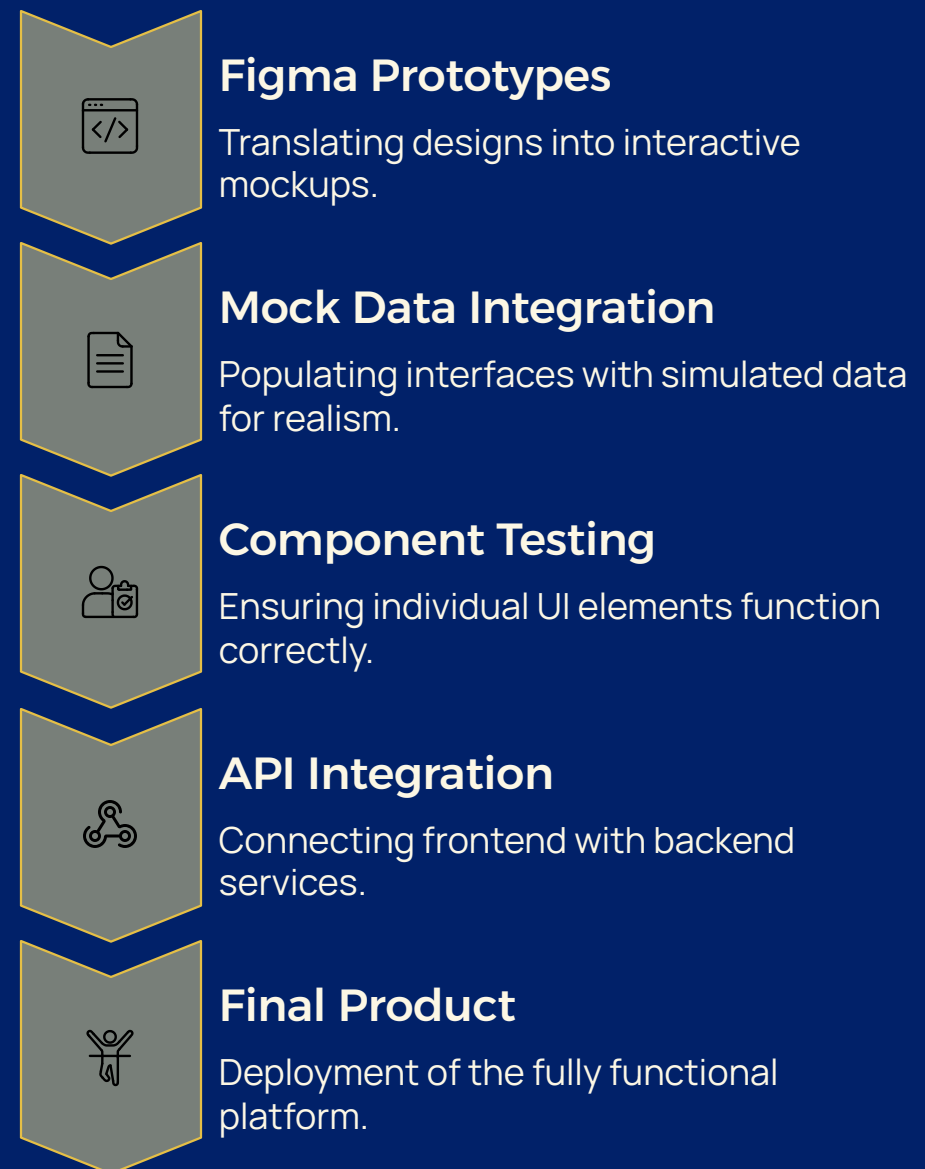
Technology Stack

We selected a powerful combination of frontend and backend technologies to ensure performance, security, and maintainability:

- Frontend: **Next.js** for server-side rendering and static site generation.
- UI Framework: **React** for building interactive user interfaces.
- Styling: **Tailwind CSS** for rapid UI development and **PostCSS** for processing CSS.
- Backend: High-performance services built with **Rust** for security and speed.
- Database: Reliable data storage using **PostgreSQL**.

Our Agile Workflow

Our development process was iterative, moving from design mockups to a fully integrated product:

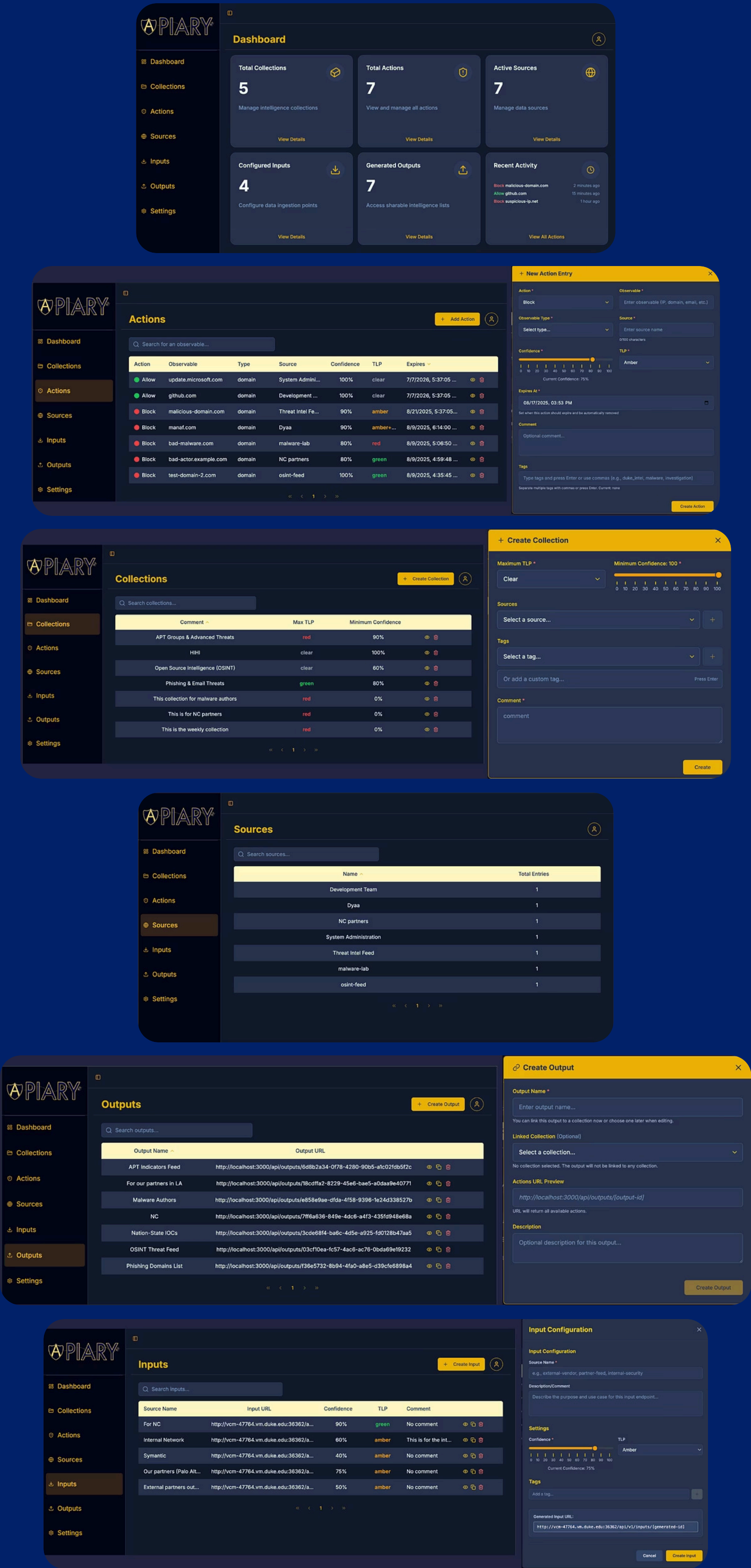


Final Product

Key Features

Each page was designed to support a specific cybersecurity workflow while maintaining consistency across the application.

- Dashboard
- Actions
- Collections
- Sources
- Outputs
- Inputs



User Workflow

- Create a security action
- Add it to a collection
- Generate an output URL
- Share intelligence with external systems

This demonstrates the application's complete end-to-end workflow, ensuring seamless threat intelligence management.

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Future Work

Next Steps

1

Expand Functionality

Integrate advanced threat intelligence feeds and automated response capabilities.

2

Improve Analytics

Develop more sophisticated data visualization and reporting tools for deeper insights.

3

Add User Roles

Introduce granular access control and roles for diverse organizational structures.

4

Open-Source Release

Prepare the platform for a community-driven open-source initiative, fostering wider adoption.

5

Support Universities

Collaborate with academic institutions to provide the platform for cybersecurity research and education.